

S.No. : 612

NBS 4101

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Following Paper ID and Roll No. to be filled in your Answer Book..

PAPER ID : 49901

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## B. Tech. Examination, 2025-26

(Odd Semester)

### MATRICES AND CALCULUS

*Time : Three Hours]*

*[Maximum Marks : 60*

**Note :-** Attempt all questions.

#### SECTION - A

1. Attempt all parts of the following :  $8 \times 1 = 8$

- Define Hermitian matrix with example.
- Define linearly dependent and independent vector.
- Find the  $n^{\text{th}}$  derivative of  $\log(ax + b)$ .

**[P. T. O.]**

(d) Define Jacobian of  $u, v$  with respect to  $x, y$ .

(e) Evaluate :

$$\int_0^a \int_0^x dy dx$$

(f) Write the value of  $\sqrt{\frac{3}{2}}$ .

(g) Define gradient of scalar point function.

(h) State Stoke's theorem.

### SECTION - B

2. Attempt any two parts of the following :  $2 \times 6 = 12$

(a) Verify the Cayley-Hamilton theorem for the matrix :

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

(b) Find the value of the  $n$ th derivative of  $y = e^{m \sin^{-1} x}$  for  $x = 0$ .

(c) Prove that :

$$2^{2n-1} \sqrt{n} \sqrt{n + \frac{1}{2}} = \sqrt{\pi} \sqrt{2n}$$

(d) Using Green's theorem, evaluate :

$$\int_C (x^2 y dx + x^2 dy)$$

where  $C$  is the boundary described counter clockwise of the triangle with vertices  $(0, 0)$ ,  $(1, 0)$ ,  $(1, 1)$ .

### SECTION - C

**Note :-** Attempt all questions. Attempt any two parts from each questions.  $8 \times 5 = 40$

3. (a) Find the rank of the matrix :

$$\begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$$

(b) Investigate for what values of  $\lambda$  and  $\mu$  the system of simultaneous equations :

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

[P. T. O.]

has (i) No solution

(ii) A unique solution

(iii) An infinite solutions

- (c) Find the eigen values and eigen vectors of the following matrix :

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

4. (a) Expand  $\sin(x/y)$  in powers of  $(x-1)$  and  $(y-\pi/2)$  as far as the terms of second degree.

- (b) Discuss the maximum and minimum of :

$$x^2 + y^2 + 6x + 12$$

- (c) If  $u = e^{xyz}$  then find the value of :

$$\frac{\partial^3 u}{\partial x \partial y \partial z}$$

5. (a) Evaluate :

$$\int_0^1 \int_0^1 e^{y/x} dy dx$$

- (b) Find the area of the ellipse :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

- (c) Find the value of  $\iiint \log(x+y+z) dx dy dz$  the integral extending over all positive and zero values of  $x, y, z$  subject to the condition  $x+y+z < 1$ .

6. (a) Show that if  $\vec{r} = \vec{a} \sin wt + \vec{b} \cos wt$  where  $\vec{a}, \vec{b}, w$  are the constant then :

$$\frac{d^2 \vec{r}}{dt^2} = -w^2 \vec{r} \text{ and } \vec{r} \times \frac{d\vec{r}}{dt} = -w (\vec{a} \times \vec{b})$$

- (b) Find the directional derivative of :

$$\phi(x, y, z) = x^2 y z + 4 x z^2$$

at  $(1, -2, 1)$  in the direction of  $2\hat{i} - \hat{j} - 2\hat{k}$ . Find the greatest rate of increase of  $\phi$ .

- (c) A vector field is given by :

$$\vec{F} = (2y+3)\hat{i} + xz\hat{j} + (yz-x)\hat{k}$$

Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  along the path  $C$  is  $x = 2t$ ,  $y = t$  and  $z = t^3$  from  $t = 0$  to  $t = 1$ .

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